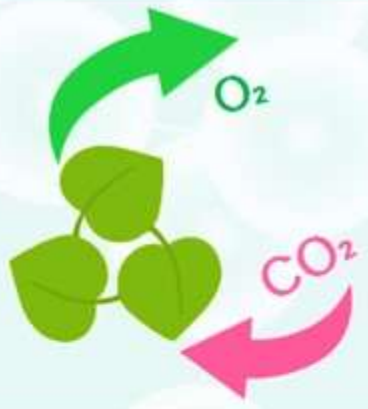


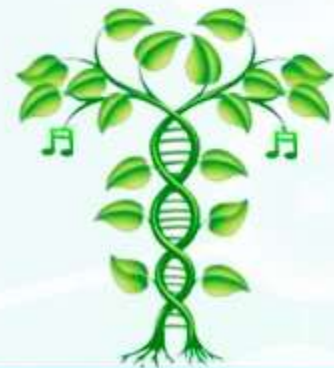


BIOLOGY



REPRODUCTION IN PLANTS-2

MODULE - 11



WHORLS OF A FLOWERING PLANT

CALYX



COROLLA



STAMEN



PISTIL



THE NON REPRODUCTIVE WHORLS

CALYX

- Usually green, sepals typically function as protection for the flower in bud, and often as support for the petals when in bloom.
- Some plants retain a thorny calyx, either dried or live, as protection for the fruit or seeds



COROLLA

- Corolla are modified leaves that surround the reproductive parts of flowers.
- They are often brightly colored or unusually shaped to attract pollinators.
- **The calyx and the corolla together make up the perianth.**



WHORLS OF A FLOWERING PLANT

CALYX



COROLLA



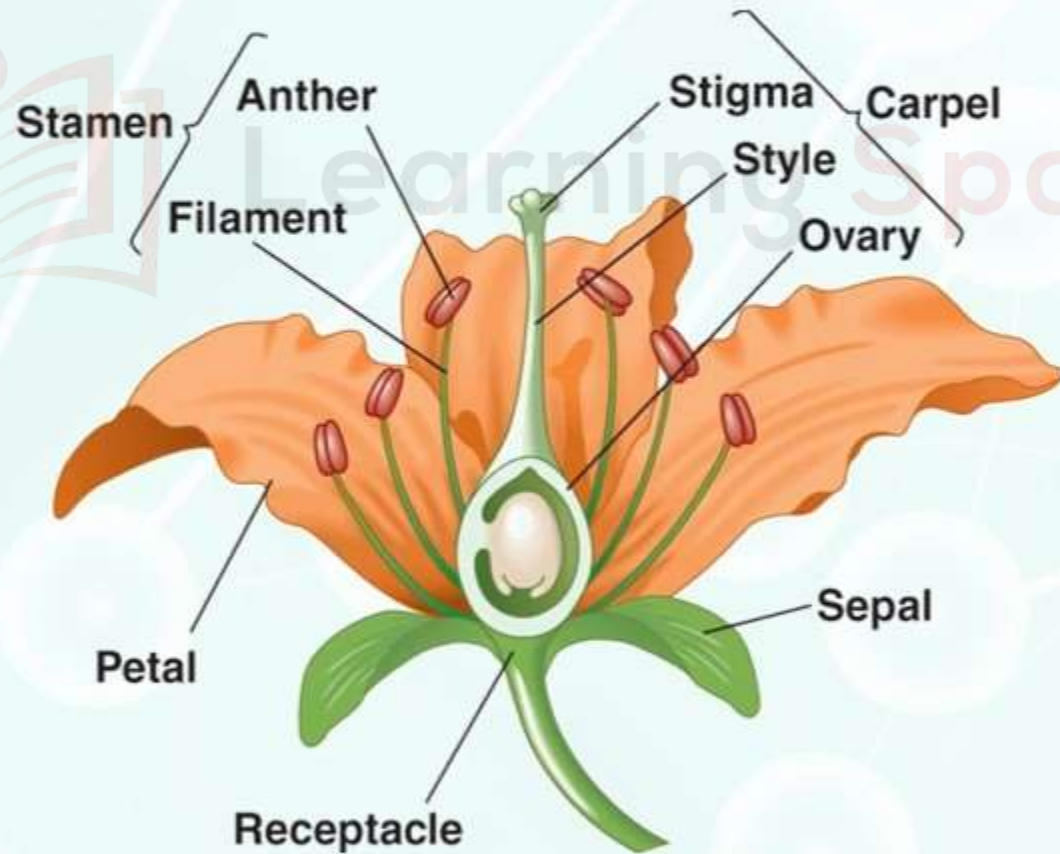
STAMEN



PISTIL

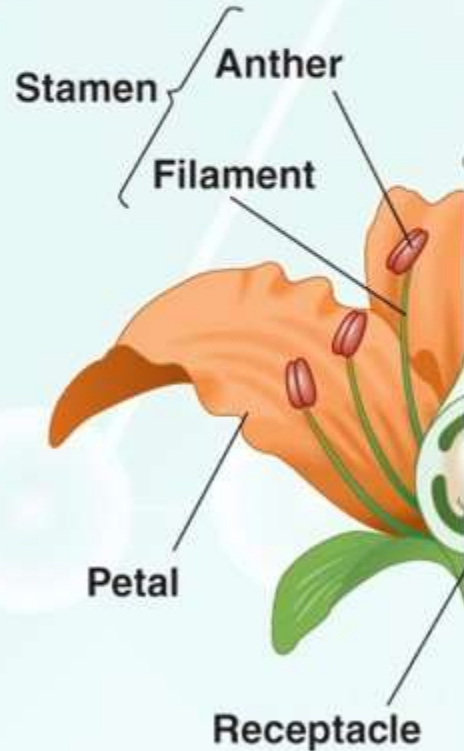


THE REPRODUCTIVE WHORLS



THE MALE REPRODUCTIVE PART

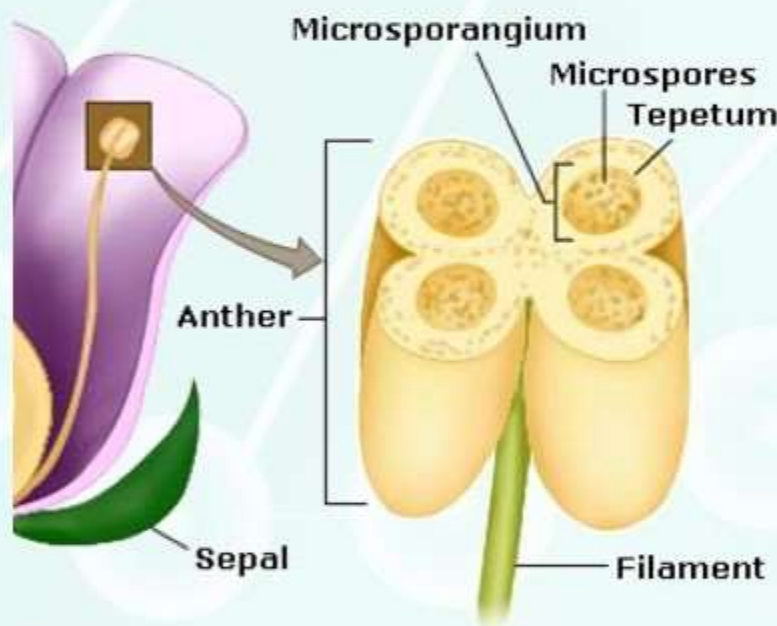
STAMEN



- A stamen typically consists of a stalk called the filament and an anther which contains microsporangia.
- The sterile tissue between the lobes is called the connective.
- The stamens in a flower are collectively called the androecium.



THE MALE GAMETE – POLLEN GRAIN

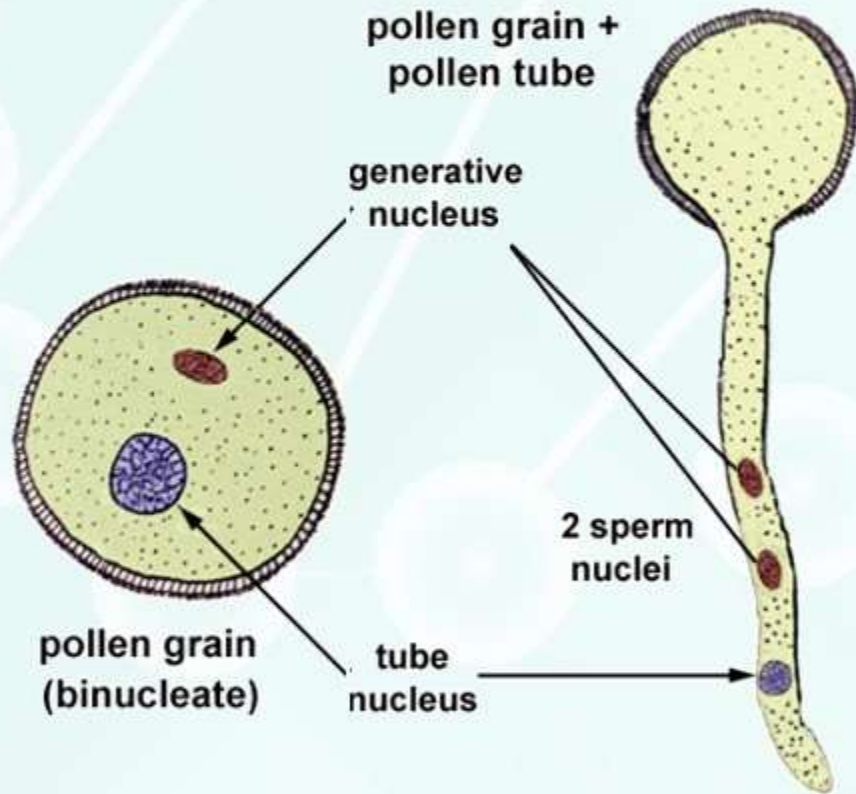


- Pollen grains contain the male gametophyte
- Pollen grains are produced by meiosis of microspore mother cells(MMC) that are located along the inner edge of the anther sacs
- The outer part of the pollen is the exine, which is composed of a complex polysaccharide, sporopollenin.
- The inner wall of the pollen grain is called intine.

The sporopollenin is chemically inert and cannot be degraded by enzymes. It is well preserved in soils and sediments.



THE MALE GAMETE – POLLEN GRAIN

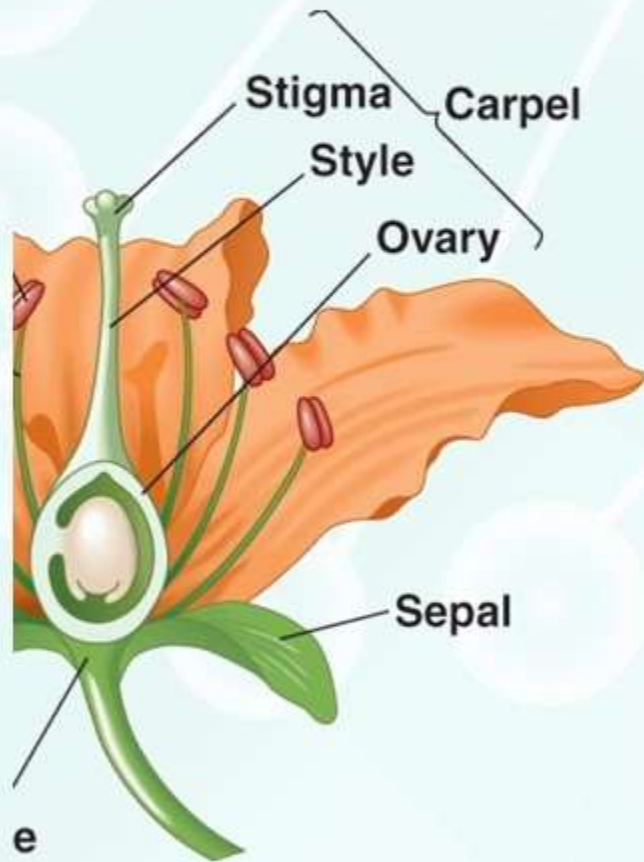


- Each pollen grain has 2 nuclei – the tube nucleus and the generative nucleus.
- The tube nucleus develops into the pollen tube.
- The generative nucleus or the germ cell divides by mitosis to produce two haploid sperm cells.
- The sperm cells reach the ovule to fuse with the female gamete.



THE FEMALE REPRODUCTIVE PART

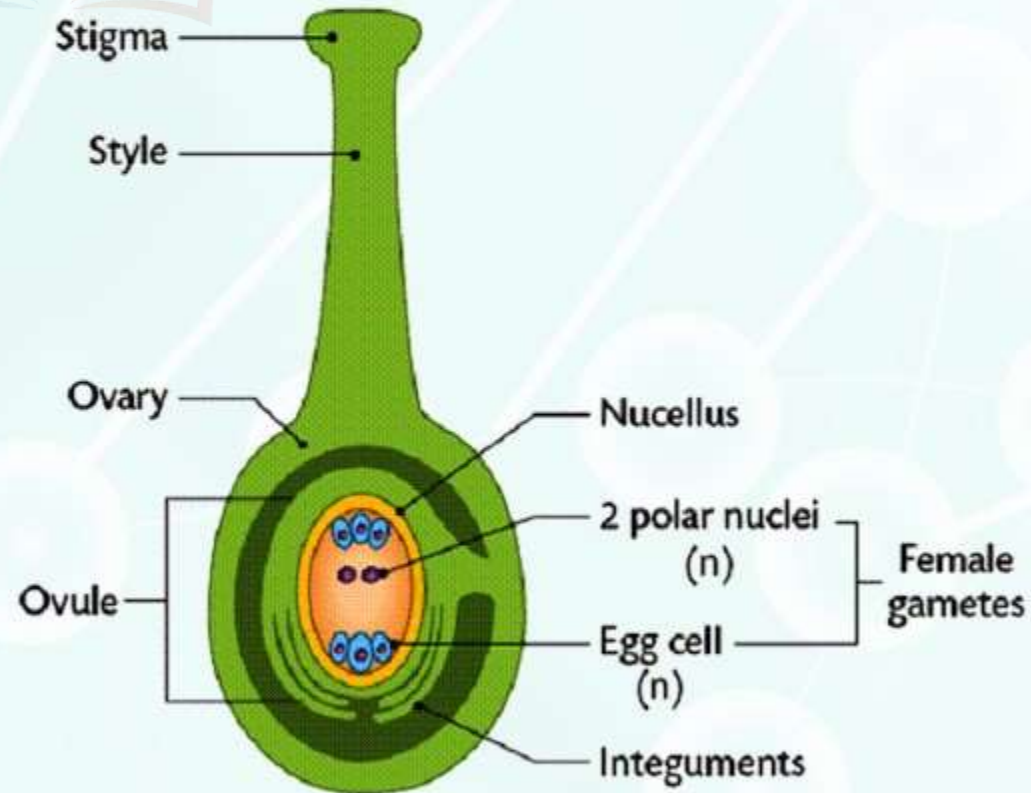
PISTIL



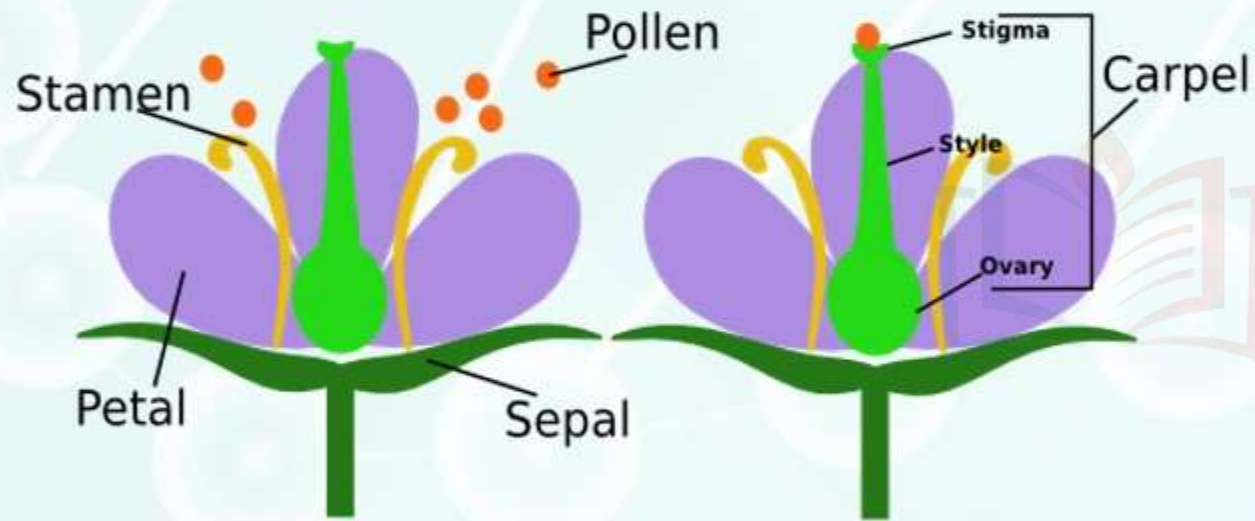
- The female reproductive part of the flower is called pistil.
- Pistil carries a type of modified leaf which bears the ovules or the egg cells called as the carpel.
- When the pistil carries only one carpel, it is said to be a simple pistil, and when the pistil carries two or more, the pistil is said to be a compound pistil.
- In the compound pistils, the carpels may remain free, or they may remain fused.
- Each carpel possesses three parts. Those are stigma, style and ovary
- The swollen base is called the **ovary**.
- a stalk called **style**, arising from the ovary; and a pollen-receptive tip called the **stigma**, variously shaped and often sticky.



THE FEMALE GAMETE – EGG CELLS



HOW THE POLLEN GRAINS ARE TRANSFERRED?

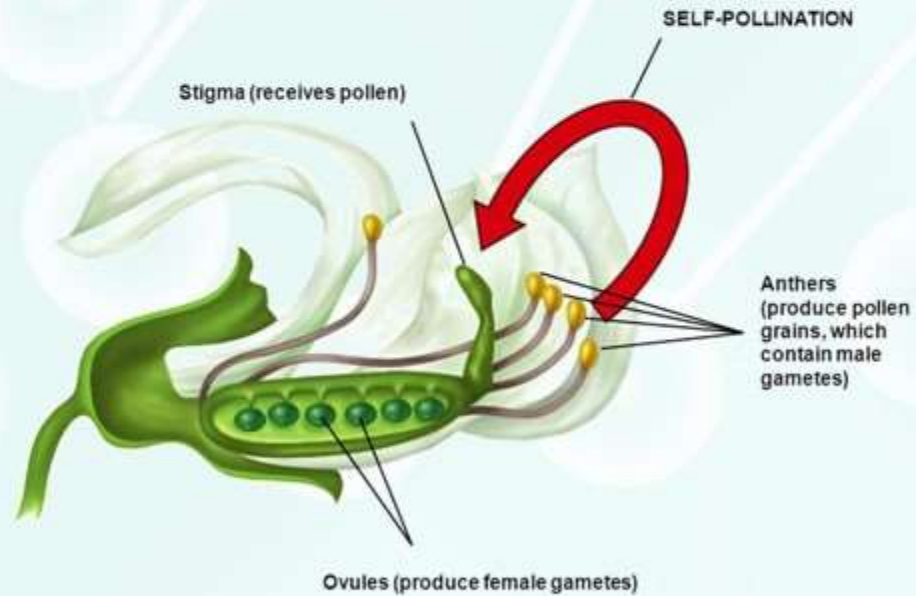


- The transfer of pollen grains from the anther of a flower to the stigma of the same flower or of a different flower is called pollination.
- Pollination helps the male and the female gamete to come closer and fertilization is facilitated.

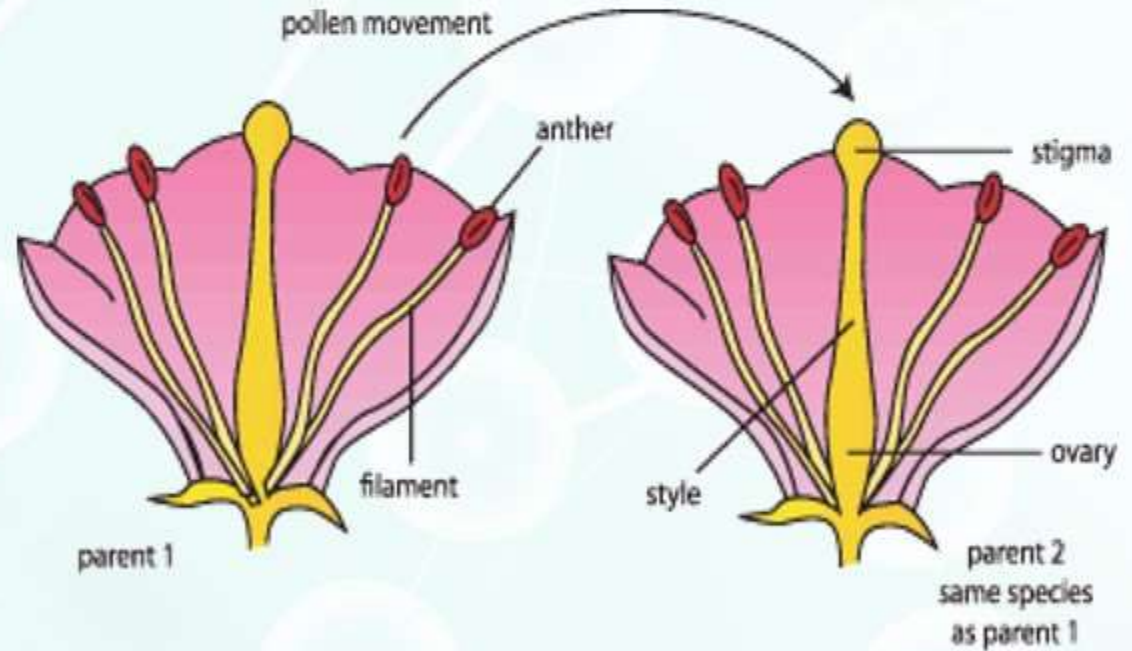


TYPES OF POLLINATION

SELF POLLINATION



CROSS POLLINATION



TYPES OF POLLINATION

SELF POLLINATION

- It is the transfer of pollen grains from anther to the stigma of the same flower or another flower born on the same plant.
- It does not require any external agency.
- **Both anthers and stigmas mature at the same time.**
- It can occur even when the flower is closed.
- Produces homozygous offsprings.
- **Does not induce any variations.**

CROSS POLLINATION

- It is the transfer of pollen grains from the anther of one flower to the stigma of another flower borne on a different plant of the same species.
- An external agent(wind, water and insects etc.) is always required.
- **Anthers and stigmas mature at different times.**
- It occurs only when the flower is open.
- Produces heterozygous offsprings.
- **Induce variations**



TYPES OF POLLINATING AGENTS

WIND

Anemophily



WATER

Hydrophily



INSECTS

Entomophily



BIRDS

Ornithophily



INDUCED POLLINATION

EMASCULATION & BAGGING

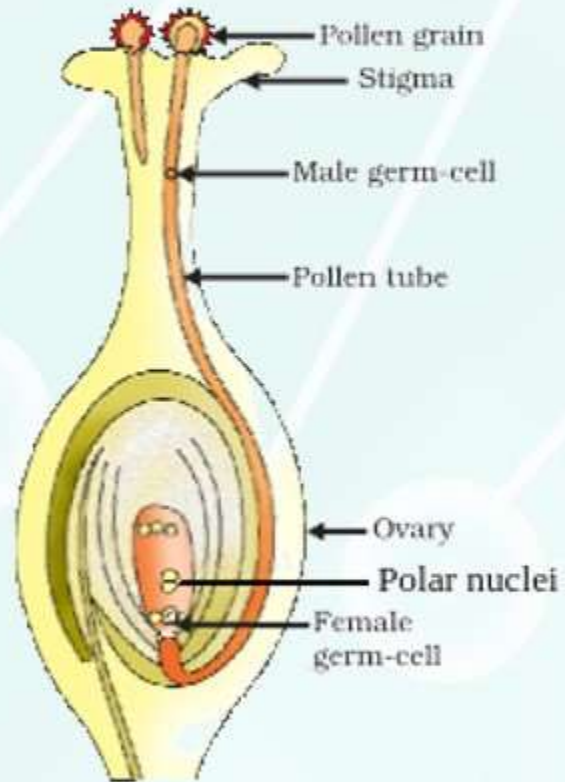


1 Emasculation is the process of **removing all the stamens from bisexual flowers** without affecting the female reproductive part (pistil), which is used in various plant hybridization techniques. **Emasculation doesn't allow self pollination** and induces variability in the seed.

2 The emasculated flower with its pistil intact is covered by a plant which is not allowed to contaminate by unwanted pollen grain. They are covered with a bag. This process is called bagging.



FERTILIZATION EVENTS



1 As the pollen tube penetrates the embryo sac, its tip penetrates in the embryo sac and reaches the egg apparatus. Ultimately, the tip of the pollen tube bursts and two male gametes are discharged.

2

- One of these male gametes fuses with the female germ cell (syngamy) causing fertilization,
- This results in a diploid zygote which gives rise to the embryo.

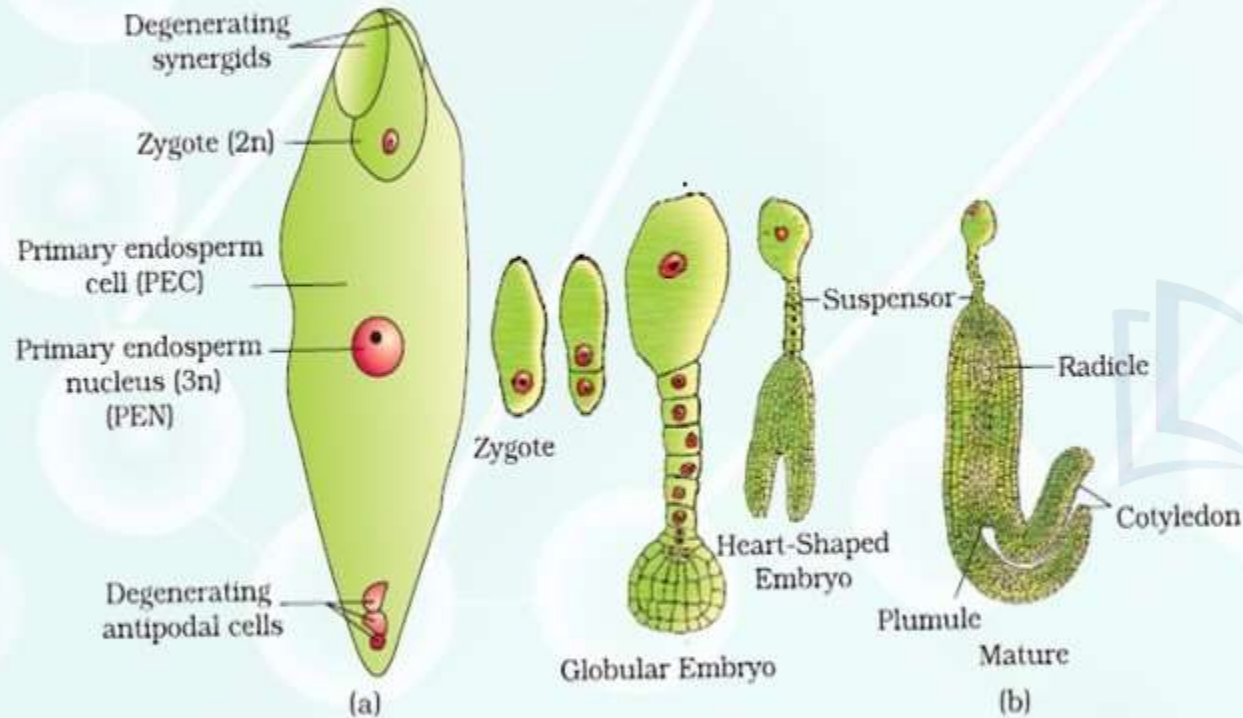
3

- The other male gamete fuses with the secondary nuclei or polar nuclei (triple fusion)
- This forms the triploid endosperm nucleus which later on gives rise to endosperm.

4 Thus, there is an event of double fertilization.



POST FERTILIZATION EVENTS

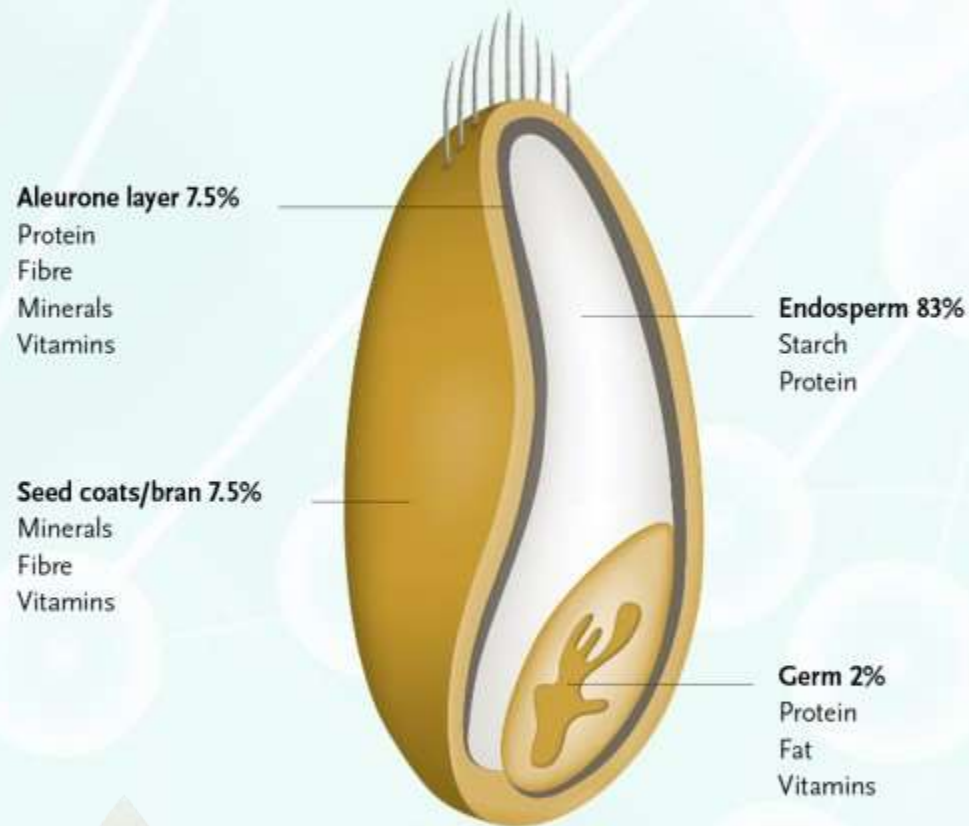


- 1 The zygote develops into the embryo of the seed with cotyledons. It contains the plumule (developing shoot) and radicle (developing root)
- 2 The endosperm nucleus divides and gives rise to the endosperm.
- 3 A fruit is formed from the ovary and the ovules become seeds.
- 4 The integuments develop into the seed coat.

Cotyledons are the primary leaf of the seed upon germination.



POST FERTILIZATION EVENTS



Anatomy of a wheat seed



TYPES OF FRUITS

TRUE FRUIT

A true fruit develops from the ovary, they are a mature ovary.



FALSE FRUIT

A false fruit or pseudocarp develops from other floral parts



PARTHENOCARPIC FRUIT

A parthenocarpic seedless fruit develops from the ovary without fertilization.



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